



PUBLIC EDUCATION GUIDE

Installing Linux on Android Phones

A safe, no-root walkthrough for running a Linux command-line environment on Android using Termux, PRoot Distro, and Debian - with public-safe cybersecurity lab notes, maintenance guidance, and practical tips.

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Release Note

This public version aligns to the GreyNOC external-release template: approved language, public-safe safety boundaries, document control, release checklist, and sanitized lab guidance.

Document Control

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Distribution	Public - free to share with attribution

Version	Date	Summary
v1.0	May 2026	Initial public education guide covering Termux, PRoot Distro, Debian, optional lab tools, maintenance, and troubleshooting.
v1.1	June 2026	Aligned to GreyNOC public-facing template; added release checklist, safety guardrails, install checklist, tips and tricks, reset/backup guidance, stronger troubleshooting, and public-safe lab workflow.

Public Release Rule	Remove internal project names, client facts, credentials, exploit walkthroughs, unreleased commitments, and private infrastructure details before publication. Keep the document educational, defensive, and permission-based.
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Plain-English Summary

This guide installs a Linux userland inside Android. Android stays in place. You get a practical terminal environment for learning Linux, scripting, SSH, Python, package management, networking basics, and authorized lab-only security tooling.

**Best-Practice
Recommendation**

Do not replace Android or flash random images to a phone unless the device vendor and the Linux distribution explicitly support that exact model. For most users, the safer path is to run Linux inside Android using Termux and PRoot Distro.

What this guide is good for

- Learning Linux commands from a phone or tablet.
- Running Python, Git, SSH, curl, whois, DNS lookups, and text editors.
- Practicing in legal home labs, classroom labs, and capture-the-flag environments.
- Using a portable shell when a full laptop is not available.

What this guide is not for

- Breaking into networks or systems without explicit authorization.
- Replacing Android with a native Linux phone operating system.
- Wireless monitor mode, packet injection, kernel-module work, or low-level driver testing.
- A guaranteed laptop replacement for all Linux or cybersecurity tasks.

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1. Scope and Responsible-Use Notice

This guide is written for public education and legitimate technical training. It shows users how to run a Linux command-line environment on Android without rooting, unlocking the bootloader, or replacing Android.

Use Linux and network tools only on devices, networks, and lab environments that you own or have explicit permission to test. Unauthorized scanning, password attacks, exploitation, interception, or attempts to bypass access controls can violate law, policy, and service terms.

GreyNOC Safety Position

GreyNOC recommends the no-root Termux approach for most Android phones. Native Linux phone installs are highly device-specific and can permanently damage a phone if the wrong image or flashing process is used.

Boundary	Public-Safe Guidance
Allowed learning	Linux basics, scripting, SSH to systems you own, DNS lookups, package management, authorized lab testing.
Needs written permission	Scanning, vulnerability validation, packet capture, password auditing, and any test against a third-party network.
Out of scope	Credential theft, stealth, persistence, evasion, bypassing access controls, real-target exploitation, or public target scanning.
Evidence habit	When practicing in a lab, keep dated notes of target ownership, commands used, and results.

2. What This Guide Installs

Termux provides an Android terminal emulator and Linux environment that works directly on Android without rooting. PRoot Distro adds a management layer for installing and logging into Linux distributions inside Termux.

Item	Recommendation
Installed approach	Termux + PRoot Distro + Debian
Why Debian?	Debian is stable, widely documented, and well supported inside PRoot environments. It is also a practical base for Parrot-style learning because Parrot OS is Debian-based.
Parrot-style use	Install only the learning and lab tools you need instead of trying to force a full desktop Parrot install onto a phone.
Android remains installed	Yes. This process does not wipe Android.
Root access	Not required.
Best install source	Use F-Droid or the official Termux project source. Avoid old, repackaged, or random APK websites.
Important Context	This is not the same as replacing Android with a full Parrot OS phone operating system. Official Parrot materials focus on desktop, virtual, container, WSL, Docker/Podman, and ARM images rather than general Android phone flashing.

3. Requirements and Pre-Install Checklist

Item	Recommendation
Android phone or tablet	A modern Android device with permission to install trusted apps. F-Droid currently lists Termux builds requiring Android 7.0 or newer.
Storage	At least 8 GB free. 20 GB or more is better if you plan to install many packages or keep logs.
Network	Stable Wi-Fi for package downloads. Avoid metered or unstable connections during install.
Power	Keep the phone charged or plugged in during installation and major upgrades.
Keyboard	Optional, but a Bluetooth keyboard makes command-line work much easier.
Screen	A larger phone, tablet, or external display improves readability for long commands.
Backups	Back up important Android files before making major changes or installing large toolsets.

Before you start

- Confirm you can install apps from trusted sources such as F-Droid.
- Make sure Android battery optimization will not immediately kill long-running package installs.
- Plan a username before Step 09. The examples use greyuser; replace it with your own username.
- Do not paste commands you do not understand. Read each command first, especially commands that delete, remove, or overwrite files.

4. Step-by-Step Installation

Step 01 - Install Termux from F-Droid

Install F-Droid first, then install Termux from the F-Droid repository. F-Droid recommends using the F-Droid client rather than downloading standalone APK files so the app can receive update notifications. Avoid old or unsupported builds from random APK websites. Open Termux after installation.

Step 02 - Grant storage access

Inside Termux, run the command below. Android will ask for permission. Tap Allow if you want Termux to access shared storage.

```
$ termux-setup-storage
```

Step 03 - Update Termux packages

Package updates reduce installation errors and pull in current dependencies.

```
$ pkg update && pkg upgrade -y
```

Step 04 - Install required Termux packages

```
$ pkg install proot-distro git wget curl nano openssh termux-services -y
```

Tip

If package downloads fail or mirrors are slow, run `termux-change-repo` from regular Termux and select a different mirror. Then retry `pkg update`.

Step 05 - Install Debian inside Termux

Debian provides the Linux base that will run inside Android. The download may take a few minutes depending on the phone and Wi-Fi connection.

```
$ proot-distro install debian
```

Step 06 - Log into Debian

```
$ proot-distro login debian
```

Step 07 - Update Debian

Run this from inside the Debian shell.

```
$ apt update && apt upgrade -y
```

Step 08 - Install everyday Linux tools

These packages support Linux learning, SSH, scripting, basic networking, manual pages, process checks, and text editing.

```
$ apt install sudo git curl wget nano vim python3 python3-pip  
dnsutils whois netcat-openbsd openssh-client iproute2  
procps man-db less tree unzip -y
```

Step 09 - Create a normal user

Working as a regular user is a better habit than always working as root. Replace greyuser with your preferred username.

```
$ adduser greyuser  
$ usermod -aG sudo greyuser
```

Step 10 - Exit and log back in as the new user

```
$ exit
$ proot-distro login debian --user greyuser
```

Step 11 - Create a quick-launch command

From regular Termux, create a small helper command so users can start Linux by typing linux-phone. Replace greyuser if you used a different username.

```
$ echo 'proot-distro login debian --user greyuser' > $PREFIX/bin/linux-phone
$ chmod +x $PREFIX/bin/linux-phone
```

After this, users can open Termux and run:

```
$ linux-phone
```

Step 12 - Confirm the environment

```
$ whoami
$ cat /etc/os-release
$ python3 --version
$ ssh -V
$ ip a
```

Good Sign

If whoami shows your normal username and /etc/os-release shows Debian, your setup is working.

5. Optional Cybersecurity Lab Tools

This optional section is for authorized labs, home networks, classroom environments, and capture-the-flag style practice. Install only the tools you need and understand what each tool does before using it.

```
$ sudo apt update
$ sudo apt install nmap tcpdump tshark nikto gobuster sqlmap -y
```

Important Limitation

Some tools may install but not work the same way they do on a laptop. Android, PRoot, and phone hardware limit raw sockets, monitor mode, packet injection, kernel modules, and privileged network operations.

Public-safe lab workflow

Phase	Good Habit
Scope	Write down the lab range, target device, owner, and allowed tests before running tools.
Start light	Begin with ping, DNS, whois, curl, and service banners before using scanners.
Log work	Save command output to dated lab notes so you can explain what happened.
Stop conditions	Stop if you see unknown systems, production assets, private data, or behavior outside the approved lab.
Clean up	Remove test accounts, close test services, and keep evidence only where allowed.

Example: save scan output in a lab

```
$ mkdir -p ~/lab-notes
$ nmap -sV 192.168.1.1 | tee ~/lab-notes/nmap-$(date +%Y%m%d).txt
```

6. Daily Use and Maintenance

Action	Command or Step
Start Linux	Open Termux and run linux-phone
Exit Linux	Run exit
Update Debian	Run sudo apt update && sudo apt upgrade -y from inside Debian
Install a package	Run sudo apt install package-name -y from inside Debian
Check IP addresses	Run ip a from inside Debian, or ip addr in Termux
Keep session awake	From Termux, run termux-wake-lock during long installs; run termux-wake-unlock when done.
Find files	Android shared storage is usually available through ~/storage after termux-setup-storage.

Quick reference session

```
$ linux-phone
$ sudo apt update && sudo apt upgrade -y
$ python3 --version
$ ssh -V
$ exit
```

Maintenance cadence

- Update Termux packages every few weeks or before a major install.
- Update Debian packages before installing new tools.
- Remove packages you no longer use to save storage.
- Keep large downloads, captures, and logs organized in a named folder.

7. Tips and Tricks

Tip	Why It Helps
Use a Bluetooth keyboard	Typing commands, arrows, Ctrl keys, and SSH sessions are much easier.
Use tmux for longer sessions	tmux lets you keep multiple terminal panes and recover from accidental disconnects. Install it with <code>sudo apt install tmux -y</code> .
Use aliases	Shortcuts reduce typing on a phone. Example: <code>alias ll="ls -lah"</code> .
Use nano first	nano is simpler for beginners than vim. Move to vim once you are comfortable.
Use tee for notes	tee saves output and still displays it on screen.
Keep lab tools minimal	Fewer tools means fewer updates, less storage use, and less confusion.
Prefer SSH for real work	Use the phone as a jump shell or learning shell; perform heavier jobs on a laptop, VM, or server when possible.
Protect secrets	Do not store API keys, passwords, client data, or private keys in shared Android storage unless encrypted and authorized.

Quality-of-life setup

```
$ sudo apt install tmux htop jq ripgrep -y
$ echo "alias ll=ls -lah" >> ~/.bashrc
$ echo "alias update=sudo apt update && sudo apt upgrade -y" >> ~/.bashrc
$ source ~/.bashrc
```

Privacy Tip

Android shared storage is convenient but more exposed than the app-private Termux directory. Keep sensitive lab notes, keys, and client material out of shared storage unless you have a clear protection plan.

8. What Will Not Work Like a Laptop

This setup does not replace Android with a native Linux phone operating system.

- Full desktop environments may be slow, battery-heavy, and difficult to use on a small screen.
- Systemd-based services may not behave the same way inside PRoot.
- Low-level network testing can be blocked by Android permissions, kernel limits, or hardware limits.
- Monitor mode, packet injection, and driver testing usually require specific chipsets, kernel support, root access, and legal authorization.
- Large compiles, data processing, and long scans can drain battery and overheat the phone.

A phone is excellent for learning commands, SSH, scripting, and light lab work, but it is not a full replacement for a dedicated Linux laptop or lab VM.

**Wireless
Research
Boundary**

For wireless research, most phone Wi-Fi chipsets do not support monitor mode or packet injection in a standard Android setup. A supported external adapter, root access, kernel support, and legal authorization are usually required. Those steps are outside the scope of this public no-root guide.

9. Troubleshooting

Symptom	Recommended Fix
pkg update fails	Check Wi-Fi, rerun pkg update, try termux-change-repo, then try pkg upgrade -y again.
F-Droid install warning	Install only from the official F-Droid site/client or official project sources. Avoid repackaged APKs.
proot-distro install fails	Confirm storage space and network stability. Remove the broken install with proot-distro remove debian, then install again.
Permission denied	Make sure the command is being run in the right environment. Use sudo only inside Debian after creating the user.
Command not found	Install the missing package or confirm whether you are in Termux or Debian.
Phone gets hot	Pause package installs, plug in the device, close other apps, and avoid long compile jobs on battery.
linux-phone fails	Check that \$PREFIX/bin/linux-phone exists, is executable, and references the username you created.
Storage not visible	Run termux-setup-storage again from Termux and approve Android storage permission.

Useful diagnostic commands

```
$ proot-distro list
$ proot-distro login debian
$ df -h
$ whoami
$ pwd
$ proot-distro remove debian
```

Delete Warning

proot-distro remove debian deletes the Debian environment. Use it only when you intentionally want to reset or repair a broken install.

10. Reset, Backup, and Uninstall

A portable phone lab should be easy to rebuild. Keep your important files separate from disposable packages and test environments.

Task	Command or Guidance
Back up home folder	From Debian, compress important notes: <code>tar -czf greynoc-linux-home-backup.tgz ~/lab-notes ~/.bashrc</code>
Export package list	Run <code>dpkg --get-selections > package-list.txt</code> inside Debian.
Remove Debian environment	From Termux, run <code>proot-distro remove debian</code> .
Reinstall Debian	Run <code>proot-distro install debian</code> , then recreate your user and packages.
Uninstall completely	Remove Debian with <code>proot-distro</code> , then uninstall Termux through Android settings if no longer needed.
Protect sensitive files	Do not upload backups containing keys, tokens, credentials, client evidence, or private network details.

Minimal backup example

```
$ mkdir -p ~/lab-notes
$ dpkg --get-selections > ~/lab-notes/package-list.txt
$ tar -czf ~/lab-notes/debian-notes-backup.tgz ~/lab-notes ~/.bashrc
```

11. Public Release Checklist

Use this checklist before publishing or sharing a public copy.

Area	Check
Classification	Footer and cover say PUBLIC / APPROVED FOR EXTERNAL RELEASE AFTER REVIEW.
Claims	Technical claims are based on public documentation or hands-on validation.
Safety	No unauthorized target instructions, stealth guidance, credential theft, persistence, or bypass language.
Privacy	No client names, private IP ranges, tokens, credentials, screenshots with personal data, or internal project names.
Commands	Commands are scoped to local setup, owned labs, or authorized training environments.
Contact	Public contact information is correct and release-safe.
Accessibility	Headings are clear, tables are readable, and code blocks are not images-only.
Review	Document owner, technical reviewer, and external communications reviewer have approved the release.
Template Alignment	This section is included to match GreyNOC public-facing release practices: sanitized content, clear owner, classification, public contact, review path, and external-safe instructions.

12. References

The following public references informed the safety notes and installation approach in this guide.

- Termux project site - <https://termux.dev/en/>
- Termux on F-Droid - <https://f-droid.org/packages/com.termux/>
- Termux help/wiki - <https://wiki.termux.com/>
- PRoot Distro project - <https://github.com/termux/proot-distro>
- Parrot OS downloads - <https://www.parrotsec.org/download/>
- Parrot OS containers docs - <https://docs.parrotsec.org/docs/containers/>
- U.S. Department of Justice cybercrime overview - <https://www.justice.gov/criminal/criminal-ccips>

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